APPM 4600 Lab 6
Variants of Newton's method for nonlinear system

1 Overview

In this lab, you will investigate techniques different techniques for building Quasi-Newton methods. In class, we focused on creating methods that eliminate the need to invert a matrix at each iteration of Newton. In this lab, you will investigate ways to improve naive Lazy Newton from class.

2 Before lab

Download and play with the Newton and Quasi-Newton codes.

Try Newton and Lazy Newton to find the root of the function

$$F(x_1, x_2) = \begin{bmatrix} x_1^2 + x_2^2 - 2 \\ e^{x_1 - 1} + x_2^2 - 2 \end{bmatrix}$$

with initial guess $\mathbf{x}_0 = (2, 0.5)$ and $(3, 5)$. Set the tolerance to 10^{-6}.

What happens? Does either method find a root? Can you pick an initial guess for which the methods will converge?

3 Lab Day: Additional Quasi-Newton methods

In this lab you will build a new Quasi-Newton method. It is a more sophisticated version of Lazy Newton ("Slacker Newton").

3.1 Slacker Newton

The advantage of Lazy Newton is that you only have to evaluate the inverse of a matrix one time for all iterations. This comes at a cost of a greater likelihood of not converging. To remedy this, an option to compute a new inverse of a Jacobian when you suspect your Lazy Newton is not converging or converging too slowly. This is what we will call *Slacker Newton*. It is not completely lazy.

3.2 Exercises: Build Slacker Newton

1. Decide on a condition under which you will think you should recompute the Jacobian. This could involve the distance between iterates, norms of a particular vector (i.e. understanding what the Newton update is), etc.
2. Implement your version of Slacker Newton. You can start with the Lazy Newton code from class.
3. Consider the vector function $\mathbf{f}(\mathbf{x}) = \begin{bmatrix} 4x_1^2 + x_2^2 - 4 \\ x_1 + x_2 - \sin(x_1 - x_2) \end{bmatrix}$. This function has roots near $(1, 0)^T$ and $(-1, 0)^T$. Apply Lazy Newton's method with an initial guess of $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Set the stopping criterion for $\epsilon = 10^{-10}$.
4. How does your Slacker Newton perform compared with your lab partner's version?
5. How does it perform for the example problem from class?

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4 Deliverables

Push your hybrid root finding method to Git and report your solutions to the questions in the exercise section on Canvas.